

Complements of Operational Research

Exercise sheet 2 — Project planning (CPM, PERT, *crashing*)

The exercises marked **Homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others are from previous tests and exams. DN and DM: normal duration and minimum duration. The cost of shortening an activity is assumed to be linear between the normal and the minimum duration.

Exercise 1 — Execution of an engineering project

Homework (1st test, 08/04/2019)

Consider the following table, which summarises the details of a project (DN and DM: normal duration and minimum duration).

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	3	10	1	20
B	A	2	25	1	30
C	A	5	40	4	55
D	B, C	7	35	5	45
E	D	7	15	5	35
F	E	2	15	1	25
G	D, F	4	20	3	45
H	G	2	30	2	30

- Determine the AON network, the possible time variation (float) of each activity, the critical path and the project duration (DP).
- Given that the client offers a bonus of 10 m.u. for each week cut from the critical path, what is the best solution for the company? Give all the justifications that allow the critical path to be shortened (*crashing*) and the company's profit to be increased.

Exercise 2 — Launching a mobile app

Homework

A company is going to launch a mobile app. The duration of each activity was estimated with three values (optimistic, most likely, pessimistic).

Activity	Description	Predecessors	t_o	t_m	t_p
A	Specification	—	2	3	4
B	Interface design	A	2	4	6
C	Server (<i>backend</i>)	A	4	6	14
D	App (<i>frontend</i>)	B	3	5	7
E	Integration	C, D	1	2	3
F	Testing	E	1	2	9
G	Marketing campaign	A	2	3	4
H	Launch	F, G	1	1	1

durations in weeks

- Build the AON network.
- Compute t_e and σ^2 for each activity.
- Determine the critical path and the expected project duration.
- What is the probability of launching the app in 20 weeks or less?
- Is there any path that, while not critical, deserves attention? Justify.

f) (Optional) Estimate the probability in d) by simulation.

Exercise 3 — Engineering project with a bonus

1st test, 13/04/2018

Consider the following table, which summarises the details of a project (execution of an engineering project).

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	4	5	2	11
B	A	2	12	2	12
C	A	4	4	3	11
D	B	4	3	1	18
E	C	5	7	3	13
F	B, E	6	8	5	14
G	D, F	3	4	2	5

- Determine the AON (*activity on node*) network and the possible time variation of each activity.
- Determine the critical path and the project duration (DP).
- Given that the client offers a bonus of 8 m.u. for each week cut from the critical path, what is the best solution for the company? Give all the justifications.

Exercise 4 — Construction project

Exam, 10/07/2018

Consider the following table, which summarises the details of a construction project.

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	4	10000	2	11000
B	A	3	6000	2	9000
C	A	2	4000	1	6000
D	B	5	14000	3	18000
E	B, C	1	9000	1	9000
F	C	3	7000	2	8000
G	E, F	4	13000	2	25000
H	D, E	4	11000	1	18000
I	G, H	6	20000	5	29000

- Determine the AON network and the possible time variation of each activity.
- Determine the critical path and the project duration (DP).
- Given that the client offers a bonus of 2000 m.u. for each week gained, what is the best solution for the company? Give all the justifications.

Exercise 5 — Project with two initial paths

Exam, 27/06/2019

Consider the following table, which summarises the details of a project.

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	3	10	1	20
B	—	8	13	7	16
C	A, B	4	2	2	6
D	A	5	10	3	12
E	B	6	20	6	20
F	C, D	4	16	1	19
G	C, E	6	5	5	7
H	F, G	8	10	3	30

- Determine the AON network and the possible time variation of each activity.
- Determine the critical path and the project duration (DP).
- Given that the client offers a bonus of 8 m.u. for each week gained, what is the best solution for the company? Give all the justifications.

Exercise 6 — Project with uncertain durations

Exam, 11/07/2019

Consider the following table, which summarises the details of a project (t_o , t_m and t_p : optimistic, normal and pessimistic time).

Activity	Predecessors	t_o	t_m	t_p
A	—	0.50	1	1.50
B	A	2	4	5
C	A	2	3	6
D	A	6	7	8
E	B	4	6	7
F	C, D	1	2	2.50
G	E, F	6	7	9
H	D	6	9	11
I	G, H	2	4	6

durations in days

- Determine the AON network.
- Determine the critical path and the project duration (DP).
- What is the probability of finishing the project in 19 days or less?
- Compute $P(17 \leq DP \leq 19)$.

Exercise 7 — Engineering project with an imposed deadline

1st test, 03/04/2024

Consider the following table, which summarises the details of a project (execution of an engineering project).

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	4	10	2	11
B	A	3	6	2	9
C	A	2	4	1	6
D	B	5	14	3	18
E	B, C	1	9	1	9
F	C	3	7	2	8
G	E, F	4	13	2	25
H	D, E	4	11	1	18
I	H, G	6	20	5	29

- Determine the AON network and the possible time variation of each activity.
- Determine all possible paths, with their durations. What is the critical path and the project duration (DP)?
- The client wants to reduce the project duration to 18 weeks. Give the activities that should be shortened and the final cost of the project, with the aim of minimising the company's expenses.

Exercise 8 — Engineering project with a reward below 32 weeks

Exam, 02/07/2024

Consider the following table, which summarises the details of a project (execution of an engineering project).

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	4	800	2	1200
B	—	8	900	6	1000
C	A	8	500	6	700
D	B	6	300	4	400
E	C, D	2	1000	1	1100
F	C	8	600	6	1000
G	E	5	200	5	200
H	F, E	3	600	2	700
I	H	9	400	7	800

- Determine the AON network and the possible time variation of each activity.
- Determine all possible paths, with their durations. What is the critical path and the project duration (DP)?
- The client wants to reduce the project duration and offers 150 m.u. for each week below 32. Give the activities that should be shortened and the final cost of the project.

Exercise 9 — Engineering project with a 36-week deadline

1st test, 27/03/2025

Consider the following table, which summarises the details of a project (execution of an engineering project).

Activity	Precedence	DN (weeks)	DN cost (m.u.)	DM (weeks)	DM cost (m.u.)
A	—	5	20	3	30
B	A	7	30	4	60
C	A	10	50	7	95
D	B, C	6	40	4	64
E	D	8	60	7	80
F	C, D	5	35	3	55
G	E, F	4	20	3	27
H	G	4	25	3	33
I	H	3	15	3	15

- Determine the AON network and the possible time variation of each activity.
- Determine all possible paths, with their durations. What is the critical path and the project duration (DP)?
- The client wants to reduce the project duration to 36 weeks. Give the activities that should be shortened and the final cost of the project, with the aim of minimising the company's expenses.